FDE INTERVIEW PREP #03

FDE INTERVIEW PREP · ISSUE 03 OF 08

Production Python and APIs.

FastAPI, Pydantic, async, SQLAlchemy, Alembic and tests.

03

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Swipe →

Questions and short answers

01 **Why Pydantic?**

Validation at trust boundaries, with clear errors.

02 **async or sync?**

async for I/O-bound work. Never block the event loop.

03 **N+1 queries?**

Eager load with selectinload, assert query counts in tests.

04 **Zero-downtime migration?**

Expand, backfill, switch reads, contract later.

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A boundary done right

PYTHON

```python
class Ticket(BaseModel):
    model_config = ConfigDict(extra="forbid")
    id: int
    priority: Literal["low", "high"]

@app.post("/tickets", status_code=201)
async def create(t: Ticket, db=Depends(session)):
    return await service.create(db, t)
```

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REST and testing follow-ups

01 **Idempotency?**

Idempotency key on POST, safe retries.

02 **Pagination?**

Cursor-based for large or changing data.

03 **What do you test?**

Business rules, bad input, and one real DB path.

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In the live round

DO

- ✓ Clarify inputs and edge cases
- ✓ Validate at the boundary
- ✓ Write one test early
- ✓ Say what you would add with time

AVOID

- ✗ Start typing immediately
- ✗ Swallow exceptions
- ✗ Ignore bad data
- ✗ Over-engineer the first version

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Practise these

- [x] CRUD API with validation
- [x] Async call with timeout + retry
- [x] Fix an N+1 query
- [x] Write an Alembic migration
- [x] Test a failure path

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KEY TAKEAWAY

Correct, validated and tested beats clever.

→ Validate at every boundary

→ Know async and the event loop

→ Ship migrations in safe steps

YOUR TURN

What do you always add to a FastAPI service before it ships?

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